

Adelowo Adebajji Oluwatimileyin

Applied Mathematician · Scientific Machine Learning Researcher

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PROFILE

M.Sc. Mathematical Engineer working at the intersection of scientific machine learning and computational PDEs: reduced-order modelling, hyper-reduction, physics-informed neural networks, operator learning, and inverse problems, validated through from-scratch numerical solvers and reproducible, held-out evaluation. Complemented by production experience developing predictive models and signal-processing pipelines for operational analytics, with hands-on skills in Python, PyTorch, NumPy/SciPy, scikit-learn, SQL, and FastAPI.

PROFESSIONAL EXPERIENCE

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|---------------------|---|
| Dec 2024 – Present | Data Scientist <i>Intuos Srl</i> , L'Aquila, Italy <ul style="list-style-type: none">Developed machine learning classifiers for real-time aircraft flight-phase identification.Designed FFT-based signal-processing pipelines for aircraft engine RPM estimation, frequency analysis, and anomaly detection.Built automated telemetry-processing pipelines and operational analytics workflows for flight-performance monitoring.Delivered REST APIs exposing machine learning models and engineering analytics to operational systems. |
| Apr 2023 – Nov 2024 | Machine Learning Engineer <i>N-and Italia Srl</i> , Italy <ul style="list-style-type: none">Developed predictive machine learning models for vending-machine delivery-failure forecasting using large-scale transaction and telemetry data.Performed exploratory and statistical analysis to identify patterns affecting system performance and operational reliability.Built data-processing and KPI-monitoring workflows to support data-driven operational decisions. |
| Jan 2023 – Mar 2023 | Operational Data Analyst <i>eResult Srl</i> , Cesena, Italy <ul style="list-style-type: none">Optimised SQL queries and automated recurring reporting workflows for business-intelligence systems.Developed reproducible data-processing workflows and transformed operational data into actionable performance metrics. |
| Apr 2022 – Oct 2022 | Digital Engineering Intern <i>Bridgestone EMIA</i> , Rome, Italy <ul style="list-style-type: none">Developed computational-geometry algorithms for engineering simulation and finite-element meshing workflows.Improved data-structure operations for tire cross-section geometry processing and implemented closest-point algorithms for cubic Bézier curves. |
| May 2022 – Aug 2022 | Machine Learning Intern <i>Fellowship.AI</i> , Remote <ul style="list-style-type: none">Implemented an InsetGAN architecture for whole-body human-image generation and developed the associated deep-learning inference workflow.Deployed model inference through FastAPI for integration with a Swift iOS application. |

SELECTED PROJECTS

Research Projects

Phase-Field Modelling of Multiphase-Flow PDEs

[GitHub repository](#)

Conservative Allen-Cahn phase-field model coupled with incompressible Navier-Stokes and an energy equation for boiling (vapour-liquid phase change); pressure Poisson equation solved directly via FFTs, extensible to 3D. Bubble-growth benchmark validated with demonstrated second-order convergence; Allen-Cahn mobility scaling validated against the boundedness criterion of Mirjalili, Ivey & Mani (2020). The 1D Stefan-problem benchmark remains an open limitation (56% error by late times, traced to the vaporisation-rate model and boundary conditions); 3D solver implemented but not yet run or validated.

ROM, Hyper-Reduction, and Neural Operators for Viscous Burgers' Equation

[GitHub repository](#)

Five-way comparison for the nonlinear 1D viscous Burgers' equation: a pseudo-spectral solver, a rank-8 POD-Galerkin ROM, a hybrid DEIM/local finite-difference hyper-reduction scheme, an MLP surrogate, and a 1D Fourier Neural Operator. The ROM is most accurate in-distribution (0.152% relative L^2 error); hyper-reduction is roughly $2\times$ faster online than the full solver, at the cost of in-distribution accuracy, a quantified tradeoff rather than a free speed-up. On an initial-condition family never seen in

training, the classical ROM generalises best; the tested FNO's ability to accept arbitrary fields did not by itself yield strong generalisation given this project's data budget.

Sparse-View Photoacoustic Tomography: Learned vs. Time-Reversal Reconstruction

[GitHub repository](#)

Synthetic acoustic forward model (j-Wave) with a compact U-Net trained to refine classical time-reversal reconstruction under 16- and 64-sensor sparse-view sensing. Learned refinement improves PSNR by 11.8 dB (16 sensors) / 10.6 dB (64 sensors) and masked SSIM from 0.188 to 0.936; a noise-robustness evaluation across five sensor-noise levels shows the improvement degrades gradually but persists even at severe (6 dB) noise.

Physics-Informed Neural Network: 1D Advection-Diffusion

[GitHub repository](#)

PyTorch PINN enforcing the PDE residual, initial condition, and periodic (value-only) boundary condition via automatic differentiation, trained without labelled interior data. Matches the analytical solution across the full space-time domain ($\sim 5 \times 10^{-3}$ relative L^2 error, independently reproduced via a full from-scratch retrain).

Differentiable PBR: Inverse Material Optimisation

[GitHub repository](#)

Pure-PyTorch differentiable renderer implementing the Cook-Torrance BRDF (GGX distribution, Schlick Fresnel, Smith geometry term), recovering spatially-varying material maps (albedo, normal, roughness, metallic) from multi-view photographs by gradient descent against 6 of 8 reference views, with the remaining 2 held out entirely. Training-view MSE 1.82×10^{-6} ; held-out-view MSE 1.02×10^{-5} , the honest measure of generalisation to an unseen viewpoint.

3D Liver Segmentation: Abdominal CT

[GitHub repository](#)

3D U-Net trained end-to-end on 131 MSD Task03 liver CT cases, with foreground-biased patch sampling, combined soft-Dice/BCE loss, cosine learning-rate annealing, and Gaussian sliding-window inference. Achieves 0.9886 Dice on a 128^3 -patch internal validation split (no independent test set).

Applied ML & Engineering Projects

Aircraft Engine Monitoring via Audio DSP

[GitHub repository](#)

Real-time engine RPM and status estimation from microphone audio alone, with no tachometer hardware: FFT-based signal processing on a Raspberry Pi, transmitting results over UART to an ESP32 controller. Validated against a real Tecnam P92 / Rotax 912 engine.

INTUOS Flight Data Monitoring Dashboard

[GitHub repository](#)

Full-stack aviation analytics platform (FastAPI + IBM DB2 backend, React/Vite frontend, Dockerised deployment) serving ML-based flight-phase and attitude classifiers (5-fold weighted F1 0.999) over recorded telemetry; developed for internal flight-performance analysis at Intuos Srl.

LLM Engineering Portfolio

[GitHub repository](#)

Nine end-to-end LLM and generative AI systems, including a Medical RAG Assistant (LangChain, ChromaDB), a two-stage Enterprise NL-to-SQL pipeline, and a pruning-and-distillation pipeline for a FinBERT sentiment classifier (96.9% accuracy/F1 at 88M params, vs. 97.6%/0.976 at the 110M baseline); LoRA/QLoRA fine-tuning applied separately to open-weight LLMs.

Additional Technical Work

Image Denoising with ResNet: Residual Channel Attention Blocks and Squeeze-and-Excitation for blind greyscale denoising; 33.71 dB PSNR / 0.918 SSIM on held-out LFW test images (28.32 dB / 0.724 for a plain-ResNet baseline without attention).

Dipole Subsurface Scattering: Differentiable dipole BSSRDF (Donner and Jensen 2005) in PyTorch, parameterised for all six Fitzpatrick skin types.

Image Inpainting: Douglas-Rachford optimisation for variational, TV-regularised image restoration.

nano-gpt: GPT-style transformer language model built from scratch in PyTorch, trained character-level on Tiny Shakespeare.

Whole-Body Diffusion Generator: IP-Adapter FaceID and SDXL face-guided portrait generation, served on serverless GPU.

Roof Segmentation: U-Net based aerial image segmentation for rooftop detection from satellite imagery.

Full portfolio with visual results: adebanjioluwatimileyin.github.io/projects.html

TECHNICAL SKILLS

Programming: Python, SQL, C++, MATLAB, JavaScript, $\text{\LaTeX}$

Machine Learning: PyTorch, TensorFlow, scikit-learn, LightGBM, supervised learning, deep learning, feature engineering, model evaluation

AI / LLM: LLMs, RAG, NLP, LangChain, OpenAI APIs, ChromaDB, NL-to-SQL, prompt engineering, embeddings, vector search

Data Science: NumPy, SciPy, Pandas, statistical analysis, exploratory data analysis, time-series and signal processing

Data Engineering / MLOps: FastAPI, Docker, MLflow, Apache Spark, Databricks, Git/GitHub, ETL/data pipelines, REST APIs

Cloud / Infrastructure: AWS, Kubernetes, Terraform, PostgreSQL, IBM DB2

Scientific Machine Learning: Reduced-order modelling (POD-Galerkin), hyper-reduction (DEIM), physics-informed neural networks, Fourier Neural Operators, inverse problems

Scientific Computing: Numerical methods, optimisation, finite-element methods, pseudo-spectral methods, computational modelling, PDEs

EDUCATION

2018 – 2021	M.Sc. Mathematical Engineering University of L'Aquila, Italy <i>Thesis:</i> Bounds on Mixing of Passive Scalars Advected by Incompressible Energy- and Enstrophy-Constrained Flows and Anomalous Dissipation. <i>Relevant coursework:</i> Partial Differential Equations, Functional Analysis, Numerical Analysis, Finite Element Methods, Scientific Computing, Optimisation, Fluid Dynamics, Calculus of Variations.
2013 – 2018	B.Sc. Mathematics Obafemi Awolowo University, Nigeria <i>Thesis:</i> Application of Galerkin Finite Element Method to the One-Dimensional Stefan Problem.

RESEARCH EXPERIENCE

2020 – 2021	M.Sc. Thesis Research University of L'Aquila Investigated mathematical bounds for passive-scalar mixing in incompressible flows. Derived analytical bounds and implemented a pseudo-spectral Python solver to validate theoretical predictions computationally.
2017 – 2018	B.Sc. Thesis Research Obafemi Awolowo University Applied the Galerkin finite-element method to the one-dimensional Stefan free-boundary problem, developing numerical algorithms for phase-boundary tracking.

AWARDS & LEADERSHIP

Awards

University of L'Aquila International Scholarship	2018 – 2021
Professor S.S. Okoya Prize for Excellence in Applied Mathematics	2018
MTN Foundation Scholarship	2014 – 2018
PTDF Scholarship	2014 – 2018
Murli T. Chellaram Foundation Scholarship	2014 – 2018

Teaching & Leadership

2016 – 2018	Founder & President, Inter-University Mathematics Club Nigeria Established a cross-university mathematics mentorship network and organised seminars on mathematics, research, and academic development.
2015 – 2018	Academic Director, NAMSS Peer Tutor Obafemi Awolowo University Coordinated workshops and tutored undergraduate students in mathematical analysis and quantitative reasoning.